AI-Sector Valuation Framework and Scenario Analysis

Strategic Balance-Sheet Competition and Adversarial Market Structure

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This paper develops an AI-sector valuation framework in which corporate value is determined not only by expected cash flows, growth, capital intensity, and discount rates, but also by strategic balance-sheet competition and adversarial market structure. The framework is motivated by two companion studies of balance-sheet adversarial attacks in equity markets: a two-firm model of leveraged balance-sheet restructuring, market impact, and liquidation costs; and a stochastic extension involving coupled price dynamics, stationary distributions, and a Nash–Fokker–Planck equilibrium. These mechanisms are incorporated as threat scenarios and valuation transmission channels rather than assumed to be universally operative or empirically established for any named company.

The analysis distinguishes fundamental enterprise value from observed equity-market value; separates ordinary competitive investment from market-price attacks; and models the effects of adversarial conduct through cash-flow impairment, financing costs, collateral constraints, market-impact distortions, and liquidity premia. It then applies the framework to the NVIDIA–Amazon–Google structure underlying the NAGMO thesis. The paper develops a scenario matrix spanning AI demand, monetisation, compute economics, capital expenditure, model commoditisation, financing conditions, and adversarial escalation. It proposes a scenario-weighted and stress-tested valuation architecture, identifies conditions under which relative-value conclusions can reverse, and specifies defensive indicators for monitoring balance-sheet and market-microstructure risk.

The central proposition is that AI-sector valuation must be conducted at two levels simultaneously: the valuation of productive AI capabilities and the valuation of each firm's resilience within a strategic financial contest. A market-price shock is not automatically a destruction of fundamental value; however, it can become economically consequential when it propagates through financing, collateral, investment, customer confidence, or corporate control channels.

The paper ends with "The End"

Keywords: artificial intelligence; equity valuation; enterprise value; strategic competition; balance-sheet adversarial attacks; market impact; liquidity; scenario analysis; NVIDIA; Amazon; Google; NAGMO.

Contents

1 Introduction **2**

2 Scope, Definitions, and Analytical Discipline **3**
2.1 The unit of valuation . . . 3
2.2 Three meanings of adversarial balance-sheet behaviour . . . 3
2.3 What the framework does not assume . . . 4

3 Fundamental AI-Sector Valuation **4**
3.1 Enterprise-value foundation . . . 4
3.2 AI-specific cash-flow architecture . . . 4
3.3 The value-chain double-counting problem . . . 5
3.4 Economic rent and bargaining power . . . 5

4 Strategic Balance-Sheet Competition **5**
4.1 The balance sheet as a competitive instrument . . . 5
4.2 Transmission channels into fundamental value . . . 5

5 Adversarial Market-Structure Risk **6**
5.1 The companion two-firm model . . . 6
5.2 A valuation distinction: market damage versus economic damage . . . 6
5.3 Execution-cost and market-depth risk . . . 7
5.4 Stochastic interaction and stationary risk . . . 7
5.5 Limits of the companion models . . . 7

6 An AI-Sector Valuation Architecture **7**
6.1 Four-layer valuation system . . . 7
6.2 Scenario-dependent enterprise value . . . 8
6.3 Adversarially robust valuation . . . 8
6.4 Risk-adjusted equity value . . . 9

7 Application to the NAGMO Universe **9**
7.1 NVIDIA: infrastructure economics and concentration risk . . . 9
7.2 Amazon: cloud scale, infrastructure commitments, and monetisation . . . 9
7.3 Google: distribution, models, cloud, and the economics of substitution . . . 10
7.4 The relative-value proposition . . . 10

8 Scenario Analysis **10**
8.1 Scenario design principles . . . 10
8.2 Scenario-specific transmission . . . 11
8.3 Adversarial escalation scenarios . . . 11
8.4 Scenario interaction matrix . . . 12

9 Valuation Stress Testing and Decision Rules **12**
9.1 Fundamental and market-structure stress tests . . . 12
9.2 Liquidity-adjusted resilience . . . 13
9.3 Break-even analysis for the NAGMO relative-value thesis . . . 13
9.4 Avoiding false neutrality . . . 14

10 Defensive Monitoring and Governance **14**
10.1 Indicators . . . 14
10.2 Governance principles . . . 14

11 Implications for AI-Sector Valuation **15**

12 Conclusion **15**

List of Figures

List of Tables

1 Coherent narratives. . . . 11
2 Scenario interaction matrix. . . . 12

1 Introduction

The valuation of artificial-intelligence companies presents a problem that conventional growth-equity analysis does not fully capture. AI firms compete over technology, compute capacity, distribution, data, talent, customer relationships, and monetisation. They also compete through capital allocation and financing decisions. Their balance sheets are therefore not passive records of past transactions: they are instruments through which firms acquire capacity, sustain pricing strategies, secure supply, finance research, make acquisitions, and withstand periods of weak cash conversion.

A further complication arises when financial-market mechanisms themselves become part of strategic competition. The companion study *Balance-Sheet Adversarial Attacks in Equity Markets* models a two-firm setting in which an initial market-capitalisation shock is followed by a balance-sheet-financed

position in the rival's equity and a subsequent liquidation whose market impact can amplify the initial damage. Its mechanism combines a price-to-book transmission assumption, Kyle-style permanent price impact, and an Almgren–Chriss execution-cost representation. The paper also considers an algorithmic execution agent and defensive surveillance measures [1].

The continuation studies repeated interaction, coupled stochastic log-price dynamics, stationary distributions, and a Nash–Fokker–Planck equilibrium. It proposes that the relationship between liquidity and attack incentives can generate a deterrence threshold and potentially destabilising retaliation dynamics [2].

These papers motivate a broader valuation question:

> What is an AI company worth when its expected productive cash flows are exposed not only to technological and commercial uncertainty, but also to strategic balance-sheet competition and adversarial financial-market behaviour?

The present paper addresses this question without presuming that every competitive balance-sheet action is an attack, or that every price dislocation changes intrinsic value. It distinguishes three domains:

1. **Productive competition:** investment, pricing, research, hiring, capacity expansion, acquisitions, and commercial execution.
2. **Strategic balance-sheet competition:** the use of debt, cash, working capital, capital expenditure, contractual commitments, asset ownership, and financing capacity to improve one's own position or constrain a rival.
3. **Adversarial market-structure conduct:** conduct that seeks to exploit trading, liquidity, collateral, financing, or price-discovery mechanisms in ways that can impair a rival's market position.

The distinction is essential. Aggressive but lawful price competition, a large capital-expenditure programme, or a strategic acquisition is not equivalent to market manipulation. The valuation framework must nevertheless account for the possibility that a rival's balance-sheet and market position can become a source of endogenous risk.

2 Scope, Definitions, and Analytical Discipline

2.1 The unit of valuation

The primary valuation object is the operating enterprise. Let firm i have fundamental enterprise value V_i^F, observed equity market value M_i, and a market-structure deviation m_i. A useful conceptual decomposition is

$$M_i = V_i^F - ND_i + N_i + m_i, \tag{1}$$

where ND_i denotes net debt and other senior claims, N_i denotes non-operating assets and adjustments, and m_i captures the difference between observed market value and the equity value implied by the fundamental valuation at the relevant time. The term m_i can be positive or negative and may include liquidity discounts, temporary price pressure, sentiment, positioning, and adversarial market effects.

This is a decomposition, not an assertion that m_i can be observed directly. It is a residual to be estimated or stress-tested.

2.2 Three meanings of adversarial balance-sheet behaviour

For valuation purposes, the following taxonomy is adopted.

Type I: Operating and strategic balance-sheet competition.
A firm uses investment, financing, cash reserves, pricing capacity, acquisitions, long-term supply contracts, or working-capital policy to strengthen its competitive position or impose economic pressure on rivals. Such conduct can be ordinary competition and may be value-creating.

Type II: Market-impact adversarial conduct.
A firm or affiliated actor seeks to exploit the relationship between financing, equity positions, trading activity, and market depth to produce a price dislocation in a rival's securities. The companion attack model is an abstract representation of this class. This paper treats it as a defensive stress scenario, not as an operational guide.

Type III: Information and capital-markets signalling risk.
Disclosure timing, guidance, capital-return decisions, financing announcements, and other signals can change investor beliefs and the market's assessment of a rival's funding needs or strategic options. Lawful disclosure and ordinary investor communication must be distinguished from deception, manipulation, or unlawful coordination.

2.3 What the framework does not assume

The analysis does not assume that NVIDIA, Amazon, Google, or any other named firm has executed a Type II attack. Nor does it infer misconduct from ordinary capital expenditure, debt issuance, acquisitions, share repurchases, or competitive pricing. Named firms are used as economic archetypes within the NAGMO thesis. Company-specific conclusions require company-specific evidence.

3 Fundamental AI-Sector Valuation

3.1 Enterprise-value foundation

For an operating AI company i, a standard unlevered discounted-cash-flow representation is

$$V_i^F = \sum_{t=1}^{T} \frac{FCFF_{i,t}}{(1+WACC_{i,t})^t} + \frac{TV_{i,T}}{(1+WACC_{i,T})^T}, \tag{2}$$

$$TV_{i,T} = \frac{FCFF_{i,T+1}}{WACC_{i,T} - g_i}, \qquad WACC_{i,T} > g_i. \tag{3}$$

The framework should use a term structure of discount rates where financing conditions and risk premia are expected to change. A single perpetual growth rate is not an adequate representation of a sector in which compute costs, model capabilities, customer adoption, and competitive pricing may change discontinuously.

Equity value is then obtained by reconciling enterprise value with net debt, preferred claims, minority interests, leases where relevant, pension or other senior obligations, and non-operating assets. The reconciliation must avoid double-counting cash, investments, or assets already represented in operating cash flows.

3.2 AI-specific cash-flow architecture

AI-sector cash flows should be decomposed into economically distinct drivers:

$$Revenue_{i,t} = \sum_{s \in \mathcal{S}_i} P_{i,s,t} Q_{i,s,t}, \tag{4}$$

$$FCFF_{i,t} = EBIT_{i,t}(1-\tau_{i,t}) + D\&A_{i,t} - CapEx_{i,t} - \Delta NWC_{i,t}. \tag{5}$$

Here $\mathcal{S}_i$ is the set of business segments, P is realised price, Q is paid quantity or utilisation, $CapEx$ is capital expenditure, and ΔNWC is the change in net working capital.

For AI businesses, the revenue and cost model should explicitly represent:

a. accelerator, networking, and systems demand;

b. cloud infrastructure consumption and utilisation;

c. model/API usage and enterprise subscriptions;

d. advertising, search, productivity, and application monetisation;

e. inference cost per task and training cost per model generation;

f. energy, cooling, land, networking, and data-centre construction;

g. depreciation, replacement cycles, and technological obsolescence;

h. customer concentration, contract duration, and cancellation or renegotiation risk.

3.3 The value-chain double-counting problem

AI firms can be both suppliers and customers of one another. A cloud provider may purchase accelerators; a model developer may purchase cloud capacity; an application provider may pay for model inference; and a platform may monetise services built on externally supplied compute.

If each transaction is capitalised as though it were independent final demand, the aggregate value of the AI ecosystem will be overstated. The analyst must distinguish:

- external final demand from intermediate transactions;
- revenue recognised by the supplier from the buyer's corresponding operating expense or capitalised asset;
- incremental economic surplus from transfers of surplus among value-chain participants.

A consolidated ecosystem view should eliminate internal transactions where the objective is to estimate total final demand. A firm-level view should retain genuine supplier revenues but explicitly model the counterparty's capacity and willingness to continue spending.

3.4 Economic rent and bargaining power

A durable AI advantage is not identical to a large revenue stream. The valuation question is whether the firm can retain economic rents after customers, suppliers, competitors, and regulators respond.

For each segment s, define an operating-surplus margin:

$$\mu_{i,s,t} = \frac{Revenue_{i,s,t} - DirectCost_{i,s,t} - IncrementalOperatingCost_{i,s,t}}{Revenue_{i,s,t}}. \tag{6}$$

A high growth rate accompanied by declining incremental margins may create less value than slower growth with durable pricing power and high returns on incremental invested capital.

4 Strategic Balance-Sheet Competition

4.1 The balance sheet as a competitive instrument

Let a firm's balance-sheet state be

$$B_{i,t} = (A_{i,t}, L_{i,t}, E_{i,t}, C_{i,t}, D_{i,t}, K_{i,t}), \tag{7}$$

where A denotes assets, L liabilities, E book equity, C available liquidity, D debt and debt-like obligations, and K productive capacity or committed capital.

The state evolves through management decisions and external shocks:

$$B_{i,t+1} = \mathcal{T}_i(B_{i,t}, u_{i,t}, u_{-i,t}, z_t, \varepsilon_{i,t}), \tag{8}$$

where $u_{i,t}$ is the firm's own strategic decision vector, $u_{-i,t}$ represents rival actions, z_t is the common macro-financial state, and $\varepsilon_{i,t}$ is an idiosyncratic shock.

The relevant strategic decisions include debt issuance or repayment, cash accumulation, capital expenditure, acquisitions, buybacks, capacity reservations, long-term customer contracts, and pricing. The same decision can be defensive, offensive, or simply operational depending on its purpose and effects.

4.2 Transmission channels into fundamental value

A balance-sheet contest affects fundamental value only when it changes one or more economic variables. The principal channels are:

1. **Cash-flow channel:** price competition, lost customers, delayed deployment, or reduced utilisation lowers expected operating cash flow.
2. **Cost-of-capital channel:** higher leverage, wider credit spreads, or reduced market confidence increases the cost of financing.

3. **Collateral channel:** a lower equity price or asset valuation weakens collateral capacity and can constrain borrowing.

4. **Investment channel:** financing stress or management distraction delays research, capacity expansion, or infrastructure deployment.

5. **Control and transaction channel:** a depressed valuation can affect acquisition currency, employee compensation, strategic partnerships, and the terms of equity issuance.

6. **Counterparty channel:** a weakened supplier, customer, cloud provider, or model partner may reduce the reliability of the firm's own revenue and operations.

A price decline that does not propagate through these channels may be a market-value shock rather than a permanent reduction in enterprise value. The distinction is central to avoiding the mistake of treating market capitalisation damage as equivalent to destroyed productive capacity.

5 Adversarial Market-Structure Risk

5.1 The companion two-firm model

The first companion paper defines market-capitalisation damage for firm i over an interval as

$$D_i(s,t) = n_i \left[S_i(s) - S_i(t) \right], \tag{9}$$

where n_i is shares outstanding and S_i is the share price. It then considers a stylised balance-sheet protocol in which a firm borrows, acquires a position in a rival's equity, and subsequently liquidates that position. Under a local price-to-book stability assumption, borrowing cash and acquiring an asset initially increases assets and liabilities together, leaving book equity unchanged. The subsequent price effect is represented through a linear permanent-impact relation [1].

The market-impact specification is

$$S_i^{post} = S_i^{pre} - \lambda_i q_i, \tag{10}$$

where q_i is signed order quantity and λ_i is a price-impact coefficient. In this representation, a smaller λ_i corresponds to greater market depth. The companion paper combines permanent and temporary execution costs in an Almgren–Chriss-style expression and derives an amplification ratio.

For the present valuation paper, the key insight is not a trading prescription. It is that a firm's observed equity value can be exposed to a strategic interaction among financing capacity, market depth, price impact, and the rival's balance-sheet response.

5.2 A valuation distinction: market damage versus economic damage

Let $P_i(t)$ denote observed equity price and $F_i(t)$ the price consistent with the market's current estimate of fundamental equity value. Define

$$m_i(t) = P_i(t) - F_i(t). \tag{11}$$

A market-impact event can change $m_i(t)$ without immediately changing $F_i(t)$. It becomes a fundamental event when the price disturbance changes financing terms, collateral, customer behaviour, employee retention, investment, or control.

A useful propagation coefficient is

$$\chi_i = \frac{\text{fundamental value impairment attributable to the event}}{\text{initial market-value disturbance}}, \tag{12}$$

interpreted as a scenario parameter rather than a directly observable constant. Values near zero describe a largely market-technical disturbance; larger values describe substantial transmission into operating and financing outcomes. The ratio can exceed one if feedback effects amplify the original shock, but such an outcome requires an explicit causal model and evidence.

5.3 Execution-cost and market-depth risk

The companion paper invokes the Almgren–Chriss framework to distinguish temporary from permanent market impact. This distinction is important for valuation:

- temporary impact may reverse after the order-flow imbalance dissipates;
- permanent impact may reflect information, changed beliefs, or persistent order-flow effects;
- financing and inventory costs can make a theoretically possible position economically infeasible;
- market depth is endogenous and may deteriorate precisely when many participants seek to exit.

The linear-impact approximation is a local modelling assumption. In real markets, price impact can be nonlinear, state-dependent, asymmetric, and affected by market-maker inventory, volatility, hidden liquidity, competing orders, trading halts, and information leakage. A valuation stress test should therefore use a range of impact functions rather than extrapolate a single linear coefficient without limit.

5.4 Stochastic interaction and stationary risk

The continuation paper models the firms' log prices as coupled mean-reverting stochastic processes and studies their stationary covariance through a Lyapunov equation. It also defines a Nash–Fokker–Planck equilibrium linking firms' strategic choices with stationary price distributions [2].

The valuation implication is that adversarial interaction can alter not only expected price levels but also:

- covariance between rival securities;
- the dispersion and tail behaviour of market-capitalisation shocks;
- the probability of repeated retaliation;
- the duration of market dislocation;
- the distribution of financing and execution costs.

The companion model's liquidity-threshold result is expressed in terms of Kyle's λ: lower λ means greater market depth. Its theorem associates sufficiently low λ with profitable attack incentives and retaliation, while higher λ is associated with a peaceful equilibrium under the stated assumptions. This is not a universal law that liquid markets are unstable. It is a model-dependent result whose applicability depends on the assumed payoff, impact, financing, and strategic-response functions.

5.5 Limits of the companion models

The two companion papers are theoretical contributions and should be interpreted with appropriate model discipline. Their conclusions depend on assumptions including local price-to-book stability, linear permanent price impact, the specified execution-cost function, feasible borrowing and position sizes, and the chosen stochastic dynamics. The first paper's reinforcement-learning comparison is reported on synthetic limit-order-book episodes; it is not evidence that a real company has used such a strategy or that the same performance would obtain in live markets. The stationary distribution and equilibrium claims in the continuation likewise depend on its stated dynamics and equilibrium conditions.

In addition, a market-capitalisation decline is not itself a cash outflow from the company. A company does not lose operating cash merely because its quoted equity price falls. The economic consequences arise through the transmission channels identified above. This paper therefore uses the companion models to motivate adversarial stress scenarios, not to treat their stylised theorem statements as empirical estimates for AI companies.

6 An AI-Sector Valuation Architecture

6.1 Four-layer valuation system

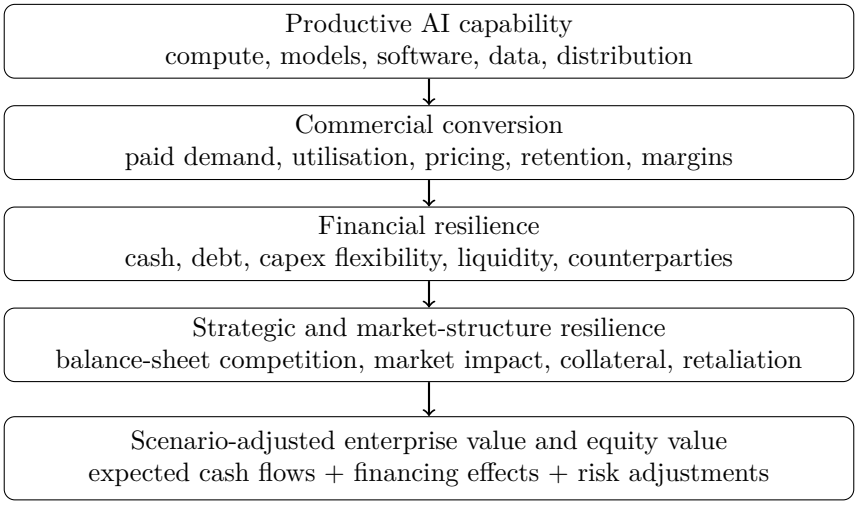
The proposed architecture has four linked layers.

Layer 1: Productive capability. Estimate the firm's technological and operational capacity: compute access, performance per unit of energy, software ecosystem, model quality, data rights, reliability, distribution, and talent.

Layer 2: Commercial conversion. Estimate paid demand, utilisation, customer retention, pricing, inference economics, gross margin, and incremental returns on capital.

Layer 3: Financial resilience. Evaluate liquidity, debt maturities, fixed commitments, capital expenditure flexibility, working capital, refinancing capacity, and counterparty concentration.

Layer 4: Strategic and market-structure resilience. Evaluate exposure to rival balance-sheet actions, market-depth shocks, collateral feedback, financing spread changes, and retaliation dynamics.



The layers are not independent. A technological advantage may require substantial capital expenditure; capital expenditure can improve productive capacity but weaken near-term liquidity; weaker liquidity can increase vulnerability to financing shocks; and a financing shock can delay the next generation of productive capacity.

6.2 Scenario-dependent enterprise value

Let $\omega \in \Omega$ index a coherent scenario, including macroeconomic conditions, AI demand, competitive responses, and adversarial events. Define

$$V_i(\omega) = \sum_{t=1}^{T} \frac{FCFF_{i,t}(\omega)}{\prod_{u=1}^{t}[1 + WACC_{i,u}(\omega)]} + \frac{TV_{i,T}(\omega)}{\prod_{u=1}^{T}[1 + WACC_{i,u}(\omega)]}. \tag{13}$$

A scenario-weighted value is

$$\mathbb{E}[V_i] = \sum_{\omega \in \Omega} p_\omega V_i(\omega), \qquad p_\omega \geq 0, \quad \sum_{\omega \in \Omega} p_\omega = 1. \tag{14}$$

Scenario probabilities should be transparent and subject to sensitivity analysis. Where probabilities are not defensible, report valuation ranges and break-even conditions rather than manufacturing precision.

6.3 Adversarially robust valuation

Let $\mathcal{A}$ denote a set of plausible adversarial and market-structure stress states. A conservative value bound is

$$V_i^{rob} = \inf_{a \in \mathcal{A}} V_i(a). \tag{15}$$

This lower envelope is not necessarily the most likely value. It is a resilience measure. A less conservative formulation uses a distributionally robust ambiguity set $\mathcal{P}$:

$$V_i^{DR} = \inf_{P \in \mathcal{P}} \mathbb{E}_P[V_i(\omega)]. \tag{16}$$

The ambiguity set should reflect uncertainty about scenario probabilities, not simply assign arbitrary worst-case outcomes to every variable simultaneously.

6.4 Risk-adjusted equity value

A useful reporting format separates operating value from financing and market-structure adjustments:

$$EV_i^{adj} = EV_i^{fund} - \mathcal{L}_i^{oper} - \mathcal{L}_i^{fin} - \mathcal{L}_i^{counterparty}, \tag{17}$$

$$Equity_i^{adj} = EV_i^{adj} - NetDebt_i + NonOperatingAssets_i - \mathcal{R}_i^{market}. \tag{18}$$

Here $\mathcal{L}^{oper}$ is expected operating impairment, $\mathcal{L}^{fin}$ is financing and refinancing impairment, $\mathcal{L}^{counterparty}$ is counterparty-related impairment, and $\mathcal{R}^{market}$ is a separately disclosed market-structure or liquidity reserve. The last term should not be deducted mechanically from intrinsic value if the underlying disturbance is purely temporary. It may instead be reported as a liquidity-adjusted equity value or stress discount.

7 Application to the NAGMO Universe

7.1 NVIDIA: infrastructure economics and concentration risk

The long NVIDIA position expresses the proposition that AI infrastructure providers can capture substantial economic rents from the expansion of compute demand. Its fundamental valuation must therefore examine:

- accelerator and systems demand across hyperscalers, enterprises, and other buyers;
- customer concentration and the dependence of demand on a relatively small number of large capital spenders;
- competitive substitution, including custom silicon and alternative accelerators;
- system-level economics, networking, software, and ecosystem durability;
- supply-chain concentration, manufacturing capacity, packaging, memory, and export restrictions;
- the risk that buyers' capital expenditure grows faster than their monetisable AI revenue.

The principal strategic vulnerability is not simply that a competitor may produce a substitute chip. It is that major customers may possess both the incentive and balance-sheet capacity to reduce dependence on a supplier, internalise parts of the stack, renegotiate economics, or shift future investment. This is a Type I strategic balance-sheet contest. It may reduce NVIDIA's future pricing power without any market manipulation.

A Type II market-structure shock would be a separate risk. It could affect the observed share price and, if persistent, the terms on which the company or its ecosystem participants raise capital. The valuation model must not equate the initial equity-price disturbance with a permanent loss of product demand.

7.2 Amazon: cloud scale, infrastructure commitments, and monetisation

The short Amazon leg in the NAGMO thesis is exposed to the proposition that cloud and service monetisation may underperform the value captured by AI infrastructure suppliers. A balanced assessment must also recognise that cloud infrastructure, distribution, enterprise relationships, and internal AI deployment can create significant option value.

The key valuation variables include:

- cloud revenue growth, utilisation, and incremental operating margins;
- AI-related capital expenditure and the speed at which new capacity becomes productive;
- depreciation schedules and the risk of economic obsolescence before accounting depreciation is complete;

- the ability to bundle AI services with existing cloud and enterprise products;
- pricing pressure and customer multi-cloud strategies;
- the resilience of the broader retail, advertising, and logistics cash-flow base.

A large investment programme may be strategically rational even when it depresses near-term free cash flow. The relevant question is whether incremental returns on invested capital exceed the firm's risk-adjusted cost of capital over the asset's economic life.

7.3 Google: distribution, models, cloud, and the economics of substitution

The short Google leg must account for the possibility that AI changes the economics of search, advertising, cloud, and productivity products. The relevant risks are not confined to model quality. They include distribution, default placement, user behaviour, advertiser returns, inference costs, regulatory constraints, and the ability to defend existing high-margin cash flows while building new businesses.

The valuation should distinguish:

- substitution of existing search activity from incremental AI usage;
- changes in query economics and advertising yield;
- the cost of serving AI-generated responses;
- cloud growth and the economics of proprietary models and infrastructure;
- model capability relative to open and closed alternatives;
- regulatory and distribution constraints.

Google may benefit from AI adoption while simultaneously facing cannibalisation of parts of its existing business model. The net effect must be modelled at the segment level rather than inferred from a single AI narrative.

7.4 The relative-value proposition

NAGMO is best treated as a relative-value hypothesis about the distribution of AI-sector economic rents. It is not enough to forecast that AI demand will grow. The thesis requires a judgement about which firms capture the surplus, when they capture it, and how much capital they must commit to do so.

For long NVIDIA and short Amazon and Google, the portfolio's fundamental scenario return can be represented as

$$R_{NAGMO}(\omega) = w_N R_N(\omega) + w_A R_A(\omega) + w_G R_G(\omega) - \mathcal{C}(\omega), \tag{19}$$

where $w_N > 0$, $w_A < 0$, $w_G < 0$, and $\mathcal{C}$ includes financing, borrow, execution, and other portfolio costs. This expression is a scenario-return identity, not a claim of beta neutrality. The original NAGMO dollar-balance construction and its reported gross coefficient sum must be analysed separately from the fundamental valuation of each issuer.

8 Scenario Analysis

8.1 Scenario design principles

The scenario set should avoid one-factor storytelling. AI demand, compute prices, capital expenditure, financing conditions, and adversarial behaviour are jointly determined. For example, a period of strong demand may coincide with aggressive capacity expansion, rising financing requirements, and eventual overcapacity. A period of weak demand may trigger price competition and strategic acquisitions while also increasing the value of cash reserves.

The following scenarios are designed as coherent narratives. They are not assigned numerical probabilities in the absence of a calibrated data set.

Scenario	AI and operating conditions	Balance-sheet / adversarial conditions	Valuation implications
S1: Infrastructure supercycle	Strong demand for accelerators, networking, and data-centre capacity; high utilisation and sustained pricing.	Hyperscalers expand capex and secure supply; financing remains available.	Supports infrastructure supplier cash flows, but tests customer capex sustainability and concentration.
S2: Monetisation catch-up	AI services generate measurable productivity gains, recurring revenue, and higher customer willingness to pay.	Cloud and platform firms fund deployment from operating cash flows and scale distribution.	Can favour cloud and application monetisers; challenges a simple hardware-over-services thesis.
S3: Compute commoditisation	Open models, efficiency gains, custom silicon, and alternative architectures reduce cost per unit of capability.	Large buyers use procurement leverage and internal investment to reduce supplier dependence.	Compresses scarcity rents; benefits efficient users and vertically integrated platforms.
S4: Capex overbuild	Capacity is installed ahead of durable demand; utilisation and pricing weaken.	Debt, leases, and long-term commitments constrain flexibility; asset impairments become plausible.	Raises depreciation and return-on-capital risk; can damage both suppliers and cloud operators.
S5: Financing and liquidity shock	AI demand remains viable, but risk premia rise and capital markets tighten.	Refinancing costs rise; collateral haircuts and market depth deteriorate.	Raises discount rates and liquidity discounts; firms with strong cash positions gain strategic optionality.
S6: Adversarial escalation	Operating fundamentals are mixed or stable, but strategic conflict intensifies.	Balance-sheet pressure, market-impact shocks, and retaliation risk increase.	Observed prices may diverge from fundamentals; financing and investment channels determine persistence.
S7: Regulatory / geopolitical fracture	Export controls, competition rules, data restrictions, or energy constraints change addressable markets.	Supply chains and capital commitments must be restructured; some assets become less fungible.	Creates asymmetric segment impairments and raises the value of geographic and supplier diversification.
S8: Broad AI disappointment	Productivity gains and willingness to pay fall short of expectations; inference costs remain high.	Capex plans are cut, contracts renegotiated, and weak projects written down.	Compresses terminal growth and multiples; cash conversion and balance-sheet resilience dominate.

Table 1: Coherent narratives.

8.2 Scenario-specific transmission

For each firm, the analyst should estimate the scenario effect on:

$$\Delta V_i(\omega) = \Delta V_i^{CF}(\omega) + \Delta V_i^{WACC}(\omega) + \Delta V_i^{TV}(\omega) + \Delta V_i^{Market}(\omega), \tag{20}$$

where the terms represent cash-flow, discount-rate, terminal-value, and market-structure effects. The market term should be reported separately from fundamental cash-flow effects to avoid double-counting.

8.3 Adversarial escalation scenarios

Adversarial escalation should be represented as a family of shocks, not a single deterministic event.

A1: Funding pressure.
A rival's financing spread widens, refinancing becomes more expensive, and discretionary investment

is delayed. The fundamental effect depends on the duration of the funding shock and the firm's available liquidity.

A2: Collateral and equity-price feedback.
A market-value decline reduces collateral capacity or increases the cost of equity issuance. The shock becomes fundamental if it constrains investment, triggers covenants, or impairs counterparties.

A3: Strategic capacity contest.
A firm commits to large capacity or long-duration contracts to secure customers or constrain a rival's access to scarce inputs. The value effect depends on utilisation, contract economics, and the opportunity cost of committed capital.

A4: Market-impact dislocation.
A severe order-flow imbalance produces a price movement that is large relative to contemporaneous fundamental news. The initial effect is classified as market-structure risk. A separate propagation model determines whether it affects financing, customer confidence, employee retention, or investment.

A5: Retaliation loop.
The affected firm responds with its own strategic or financial action. The resulting game may increase volatility, correlation, and tail risk. The continuation paper's coupled stochastic framework motivates modelling such feedback, but its specific equilibrium properties must be independently validated before calibration.

8.4 Scenario interaction matrix

The following matrix indicates the expected direction of first-order pressure, not a forecast of share-price returns.

Table 2: Scenario interaction matrix.

Scenario	NVIDIA	Amazon	Google
Infrastructure supercycle	Strong positive	Mixed / positive	Mixed / positive
Monetisation catch-up	Positive, but relative share uncertain	Positive	Positive
Compute commoditisation	Negative rent pressure	Potentially positive	Potentially positive
Capex overbuild	Negative demand and inventory risk	Negative utilisation / return risk	Negative utilisation / return risk
Financing and liquidity shock	Negative multiple pressure	Relative resilience depends on cash flows	Relative resilience depends on cash flows
Adversarial escalation	Firm-specific and path-dependent	Firm-specific and path-dependent	Firm-specific and path-dependent
Regulatory / geopolitical fracture	Supply and market-access risk	Cloud and platform constraints	Distribution and platform constraints
Broad AI disappointment	High expectation-reset risk	AI capex and monetisation reset	Search / cloud / monetisation reset

9 Valuation Stress Testing and Decision Rules

9.1 Fundamental and market-structure stress tests

A defensible valuation process should run two separate stress-test families.

Fundamental stress tests:

- revenue growth and customer adoption;
- realised price and gross margin;

- compute utilisation and inference cost;
- capital expenditure, depreciation, and replacement cycles;
- customer concentration and contract renewal;
- terminal growth and return on invested capital.

Market-structure stress tests:

- bid–ask spread and market-depth deterioration;
- temporary and permanent price-impact ranges;
- securities-lending availability and borrow-cost shocks;
- financing spread and margin requirement changes;
- correlated liquidation and execution delays;
- market-price dislocation and recovery duration.

The two families should be linked only through explicit transmission variables. This prevents a market-price shock from being counted once as a multiple contraction and again as a permanent cash-flow impairment without a causal basis.

9.2 Liquidity-adjusted resilience

Let L_i be a firm's available liquidity, O_i its committed cash obligations over a stress horizon, and F_i its stressed operating cash generation. A simple liquidity coverage measure is

$$LC_i = \frac{L_i + F_i}{O_i}. \tag{21}$$

This measure should be supplemented by maturity ladders, committed credit facilities, restricted cash, lease obligations, and the timing of capital expenditure. A high cash balance is not sufficient if obligations are concentrated or cash is geographically or legally restricted.

At portfolio level, a separate liquidation-resilience measure is required:

$$LR_p(H) = \frac{\text{estimated executable position value over horizon } H}{\text{gross portfolio exposure}}. \tag{22}$$

The measure should be computed under normal and stressed depth assumptions and should account for the short side independently. It is not equivalent to the issuer's balance-sheet liquidity.

9.3 Break-even analysis for the NAGMO relative-value thesis

The relative-value thesis should be tested through explicit break-even conditions. Let V_N, V_A, and V_G denote scenario-adjusted equity values, and let P_N, P_A, and P_G denote market prices. A relative-value spread can be represented as

$$\Pi(\omega) = w_N \left(\frac{V_N(\omega)}{P_N} - 1 \right) + w_A \left(\frac{V_A(\omega)}{P_A} - 1 \right) + w_G \left(\frac{V_G(\omega)}{P_G} - 1 \right) - \mathcal{C}(\omega). \tag{23}$$

The weights must be normalised consistently with the portfolio's actual dollar exposures. The investment proposition is supported only when the expected or robust spread remains positive after borrow costs, financing, execution, and plausible scenario losses.

The break-even surface is

$$\mathbb{E}[\Pi(\omega)] = 0. \tag{24}$$

A more conservative decision rule requires

$$\inf_{P \in \mathcal{P}} \mathbb{E}_P[\Pi(\omega)] > 0, \tag{25}$$

where $\mathcal{P}$ is a defensible ambiguity set. This is a robustness criterion, not a guarantee of profit.

9.4 Avoiding false neutrality

Dollar neutrality, beta neutrality, factor neutrality, and adversarial resilience are distinct properties. A balanced long–short book can still have material exposure to AI capex, growth duration, semiconductor cyclicality, interest rates, or common liquidity shocks. Nor does a balanced book necessarily remain balanced after a market dislocation.

The valuation framework therefore reports:

1. net dollar exposure;
2. gross exposure relative to committed capital;
3. broad-market and style-factor sensitivities;
4. AI infrastructure and monetisation sensitivities;
5. financing and securities-borrow costs;
6. scenario-specific market-depth and liquidation risk.

10 Defensive Monitoring and Governance

10.1 Indicators

The following indicators can support defensive monitoring. They are screening variables, not proof of misconduct.

Indicator	What to monitor	Valuation relevance
Balance-sheet discontinuity	Unusual debt, asset, or investment changes relative to operating plans and disclosed uses.	May signal a change in funding capacity or risk appetite.
Capital commitment intensity	CapEx, leases, capacity reservations, purchase commitments, and cancellation terms.	Determines fixed-cost burden and flexibility.
Working-capital divergence	Receivables, inventory, payables, contract assets, and cash conversion relative to revenue.	Can reveal pressure on cash generation or demand quality.
Borrow and collateral conditions	Borrow availability, financing spreads, margin terms, and collateral haircuts where observable.	Indicates short-side and refinancing vulnerability.
Market-depth instability	Spreads, depth, volatility, order-flow imbalance, and price response to volume.	Helps distinguish market dislocation from fundamental repricing.
Counterparty concentration	Dependence on a small number of suppliers, customers, cloud partners, or financing counterparties.	Measures transmission and contagion risk.
Disclosure and event timing	Material announcements, financing events, and unusual price/volume behaviour.	Supports event attribution; does not establish intent.

10.2 Governance principles

A robust process should:

- separate public information, analyst inference, and unverified allegations;
- preserve an auditable record of scenario assumptions and probability weights;
- require independent review of unusually large or opaque related-party exposures;
- distinguish ordinary competition from potentially manipulative conduct;

- avoid trading or corporate decisions based solely on uncorroborated market signals;
- include legal, compliance, and market-conduct review where relevant;
- define liquidity and leverage limits before a stress event rather than during one.

The objective is resilience and accurate valuation, not replication of adversarial tactics. Market surveillance should identify suspicious patterns for appropriate review, not label a firm or transaction as abusive solely because it is large, unusual, or strategically consequential.

11 Implications for AI-Sector Valuation

The framework yields several propositions.

Proposition 1: AI growth is not equivalent to AI rent capture. Aggregate demand can expand while the distribution of value shifts among chip suppliers, cloud platforms, model providers, application firms, and customers.

Proposition 2: Capital expenditure is both an asset and a strategic commitment. Its value depends on utilisation, economic life, energy and operating costs, and the return on incremental capital. The same expenditure can strengthen a firm's position while weakening its near-term liquidity.

Proposition 3: Balance-sheet strength has option value. Cash, low refinancing dependence, flexible investment schedules, and diversified counterparties can allow a firm to withstand competitive escalation and invest when rivals are constrained.

Proposition 4: Market-capitalisation damage is not automatically fundamental damage. It becomes fundamental when it transmits into financing, collateral, investment, operations, customer behaviour, or control.

Proposition 5: Liquidity has a dual character. Market depth normally improves execution and price discovery, but under particular adversarial payoff structures it may also reduce the cost of imposing market impact. The net effect depends on strategic incentives, market design, and constraints; it is not universally destabilising.

Proposition 6: Relative-value conclusions are scenario-dependent. A thesis that favours infrastructure suppliers over cloud monetisers can reverse under compute commoditisation, customer vertical integration, improved cloud monetisation, capex overbuild, or a shift in the distribution of AI rents.

12 Conclusion

AI-sector valuation must move beyond a single growth narrative. It must identify where economic rents are generated, who captures them, how much capital is required to sustain them, and how resilient the resulting cash flows are to strategic competition.

The two companion studies of balance-sheet adversarial attacks provide a useful conceptual warning: equity-market structure and corporate financing decisions can interact in ways that make observed prices vulnerable to strategic shocks. Their stylised two-firm mechanism and stochastic extension motivate the inclusion of market-impact, financing, liquidity, and retaliation scenarios in AI-sector valuation. Their theoretical and synthetic results should not, however, be mistaken for evidence of actual attacks by any named AI company or for calibrated estimates of real-world attack costs.

For the NAGMO thesis, the key question is not merely whether NVIDIA is a superior AI business to Amazon or Google. It is whether the expected distribution of AI-sector economic rents, after accounting for capital intensity, customer bargaining power, compute substitution, monetisation, balance-sheet resilience, and adversarial risk, favours the long–short construction at its actual entry prices and after all portfolio costs.

The appropriate valuation conclusion is therefore conditional. NVIDIA may capture infrastructure scarcity rents; Amazon and Google may capture substantial value through cloud, distribution, and monetisation; and all three may be exposed to strategic balance-sheet competition. The relative outcome depends on the scenario, the time horizon, and the path through which market shocks propagate into fundamentals.

The proposed framework treats adversarial conduct as a source of endogenous risk without allowing it to displace the central task of valuation: estimating sustainable, risk-adjusted economic cash flows. Its governing distinction is:

$$\boxed{\text{Observed Market Value} \neq \text{Fundamental Enterprise Value}} \tag{26}$$

The difference can be temporary, persistent, or economically consequential. Determining which case applies is the central analytical task.

References

[1] Ghosh, S. (2026). *Balance-Sheet Adversarial Attacks in Equity Markets: Amplified Retaliation, Market-Impact Leverage, and an Algorithmic Framework*. Unpublished working paper supplied for this study.

[2] Ghosh, S. (2026). *Balance-Sheet Adversarial Attacks in Equity Markets (Continued): Amplified Retaliation, Stochastic Equilibrium, and Stationary Distributions*. Unpublished working paper supplied for this study.

[3] Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6), 1315–1335.

[4] Almgren, R. and Chriss, N. (2001). Optimal execution of portfolio transactions. *Journal of Risk*, 3(2), 5–39.

[5] Amihud, Y. (2002). Illiquidity and stock returns: cross-section and time-series effects. *Journal of Financial Markets*, 5(1), 31–56.

[6] Bertsimas, D. and Lo, A. W. (1998). Optimal control of execution costs. *Journal of Financial Markets*, 1(1), 1–50.

[7] Gatheral, J. (2010). No-dynamic-arbitrage and market impact. *Quantitative Finance*, 10(7), 749–759.

[8] Huberman, G. and Stanzl, W. (2004). Price manipulation and quasi-arbitrage. *Econometrica*, 72(4), 1247–1275.

[9] Damodaran, A. (2012). *Investment Valuation: Tools and Techniques for Determining the Value of Any Asset*. 3rd ed. Wiley.

[10] Koller, T., Goedhart, M., and Wessels, D. (2020). *Valuation: Measuring and Managing the Value of Companies*. 7th ed. Wiley.

[11] Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. (2017). Proximal policy optimization algorithms. arXiv:1707.06347.

[12] Gardiner, C. W. (2004). *Handbook of Stochastic Methods*. 3rd ed. Springer.

The End